

$$\frac{dy}{dx} = -2y \cos(x)$$

Identity $y' = \lambda(x)y$

$$\lambda(x) = -2\cos(x)$$

Exact Solution $\int \frac{1}{y} dy = \int -2\cos(x) dx$

$$\ln|y| = -2\sin(x) + C$$

$$y(x) = Ce^{-2\sin(x)}$$

$$y(x) = y_0 e^{-2\sin(x)}$$

$$y_0 e^{-2} \leq y(x) \leq y_0 e^2$$

Euler's Method
Update

$$y_{n+1} = y_n + hf(x_n, y_n)$$

$$y_{n+1} = y_n + h(-2y_n \cos(x_n))$$

$$y_{n+1} = (1 - 2h\cos(x_n))y_n$$

$$G(x_n) = 1 - 2h\cos(x_n)$$

Stability Condition

$$|G(x_n)| \leq 1$$

$$|1 - 2h\cos(x)| \leq 1$$

Case 1: $\cos(x) > 0$ $0 \leq h\cos(x) \leq 1$ $h \leq \frac{1}{\cos(x)}$ $h \leq 1$

Case 2: $\cos(x) = 0$ $G = 1$

Case 3: $\cos(x) < 0$ $G = 1 + 2h|\cos(x)| > 0$ $|G| > 1$

Conclusion: $|1 - 2h\cos(x)| \leq 1$, but there is no single positive step size h that guarantees stability for all x .

- where $\cos(x) > 0$, euler is stable if $h \leq 1$

- where $\cos(x) < 0$, euler is unstable for any $h > 0$

∴ This shows that euler's method is conditionally stable. This oscillatory nature of $\cos(x)$ causes alternating stable and unstable regions. Therefore, a small stepsize h is required for accuracy.